<u>SOLUTIONS TUTORIAL SHEET - 1</u>

<u>(DIGITAL CIRCUITS AND SYSTEMS)</u>

A1. (i)

	Quotient	Remainder
3287/8	410	7
410/8	51	2
51/8	6	3
6/8	0	6

$(3287)_{10} = (6327)_8$

0.5100098 x 8 4.0800784	0.0800784 x 8 0.6406272	0.6406272 x 8 5.1250176	0.1250176 x 8 1.0001408
4	**0**	**5**	**1**

$(0.5100098)_{10} = (0.4051)_8$

$$\mathbf{(3287.5100098)_{10} = (6327.\ 4051)_8}$$

Now write binary equivalent of each digit i.e

6 : 110
3 : 011
2 : 010
7 : 111
4 : 100
0 : 000
5 : 101
1 : 001

$$\mathbf{(3287.5100098)_{10}} = \mathbf{(6327.\ 4051)_8} = \mathbf{(}110\mathbf{011}010\mathbf{111.}\ 100\mathbf{000}101\mathbf{001}\mathbf{)_2}$$

(ii)

	Quotient	Remainder
675/16	42	3
42/16	2	10=A
2/16	0	2

$(675)_{10} = (2A3)_{16}$

0.625 x 16 = 10.0 i.e A

$(0.625)_{10} = (0.A)_{16}$

$(675.625)_{10} = (\ 2A3.A\ )_{16}$

Now write binary equivalent of each digit i.e

2 : 0010

A : 1010

3 : 0011

$(675.625)_{10} = (\ 2A3.A\ )_{16} = (001010100011.\ 1010)_2$

A2.

(i) using r's complement i.e 16's complement :

16's complement of $-(974B)_H = (68B5)_H$

$$
\begin{array}{r}
F\ F\ F\ F \\
-\ 9\ 7\ 4\ B \\
\hline
6\ 8\ B\ 4 \\
+\qquad 1 \\
\hline
6\ 8\ B\ 5 \\
\hline
\end{array}
$$

$(587C)_H + (68B5)_H = ?$

$$
\begin{array}{r}
5\,8\,7\,C \\
+\,6\,8\,B\,5 \\
\hline
\_\_C\,1\,3\,1
\end{array}
$$

If no carry, take r's complement & put negative sign

$-(3ECF)_H$

$-(3ECF)_H =$ **$-(16079)_{10}$**

$(16^3 \times 3) + (16^2 \times E) + (16^1 \times C) + (16^0 \times F) = (16079)_{10}$

Now using (r-1)'s complement i.e 15's complement :

15's complement of $-(974B)_H = (68B4)_H$

$$
\begin{array}{r}
5\,8\,7\,C \\
+\,6\,8\,B\,4 \\
\hline
\_\_C\,1\,3\,0
\end{array}
$$

If no carry, take r's complement

$(3ECF)_H$

(ii) using r's complement i.e 2's complement :

$$\begin{array}{r} (0111)_2 \\ -(1101)_2 \\ \hline (\quad\quad)_2 \\ \hline \end{array} \xrightarrow{\text{2's Complement}} \begin{array}{r} (0111)_2 \\ +(0011)_2 \\ \hline 1010 \end{array}$$

finding 2's complement :

$$\begin{array}{r} \mathbf{1}\,\mathbf{1}\,\mathbf{1} \\ -\ \mathbf{1}\,0\,1\,0 \\ \hline 0110 \\ \hline \end{array}$$

if no carry, find 2's complement & put negative sign

final answer : $\mathbf{(-\ 0110)_2 = (06)_8}$

Now using (r-1)'s complement i.e 1's complement :

$$\begin{array}{r} (0111)_2 \\ -(1101)_2 \\ \hline (\quad\quad)_2 \\ \hline \end{array} \xrightarrow{\text{1's Complement}} \begin{array}{r} (0111)_2 \\ +(0010)_2 \\ \hline 1001 \end{array}$$

finding 1's complement :

$$\begin{array}{r} \mathbf{1}\,\mathbf{1}\,\mathbf{1}\,\mathbf{1} \\ -\ \mathbf{1}\,0\,0\,1 \\ \hline 0110 \\ \hline \end{array}$$

if no carry, find 1's complement & put negative sign

final answer : $\mathbf{(-\ 0110)_2 = (06)_8}$

(iii)

using r's complement i.e 8's complement :

$$
\begin{array}{r}
(564)_8 \\
-(3703)_8 \\
\hline
(\quad)_2 \\
\hline
\end{array}
\xrightarrow{\text{8's Complement}}
\begin{array}{r}
(564)_8 \\
+(5075)_8 \\
\hline
5661
\end{array}
$$

finding 8's complement :

$$
\begin{array}{r}
\mathbf{7\,7\,7\,8} \\
-\ \mathbf{3\,7\,0\,3} \\
\hline
5\,0\,7\,5 \\
\hline
\end{array}
$$

if no carry, find 8's complement & put negative sign

final answer : -(2117)$_{\mathbf{8}}$ **= -(010001001111)$_{\mathbf{2}}$**

Now using (r-1)'s complement i.e 7's complement :

$$
\begin{array}{r}
(564)_8 \\
-(3703)_8 \\
\hline
(\quad)_2 \\
\hline
\end{array}
\xrightarrow{\text{7's Complement}}
\begin{array}{r}
(564)_2 \\
+(5074)_2 \\
\hline
5660
\end{array}
$$

finding 7's complement :

$$
\begin{array}{r}
\mathbf{7\,7\,7\,7} \\
-\ 3\,7\,0\,3 \\
\hline
5\,0\,7\,4 \\
\hline
\end{array}
$$

if no carry, find 7's complement & put negative sign

final answer : -(2117)$_{\mathbf{8}}$ **= -(010001001111)$_{\mathbf{2}}$**

A3. **Let the base is x then**

$$(24)_x + (14)_x = (41)_x$$
$$(2x^2 + 4x) + (x^2 + 4x) = 4x^2 + x$$
$$3x^{2+}8x = 4x^2 + x$$
$$x^2 = 7x$$
$$x = 7$$

A4.

(i)

$$(\sqrt{41})^2 = (5)^2$$
$$(41)_B = (25)_{10}$$
$$(4B + 1)_{10} = (25)_{10}$$
$$B = 6$$

(ii) $(E0B)_H - (ABF)_H = (34C)_H$

$$(34C)_H = (001101001100)_2 = (1514)_8$$

7's complement of $(1514)_8 = (6263)_8$

8's complement of $(1514)_8 = (6263)_8 + 1 = (6264)_8$

A5.

(i) **Value in register :** 100010010111

(a) **In BCD Code :** 1000 1001 0111

Decimal digits : 8 9 7 i.e answer 897

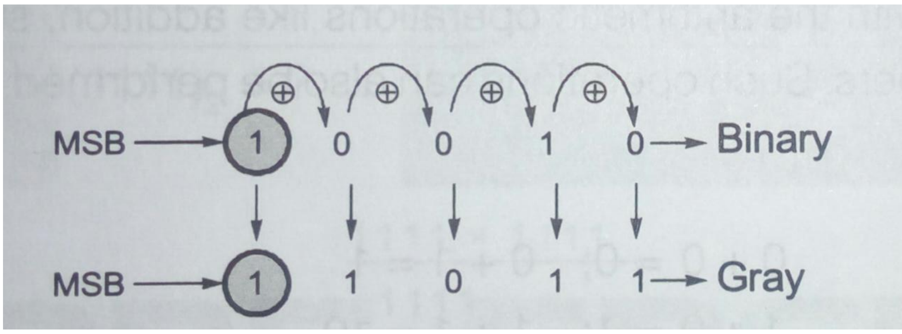
(b) **In Excess-3 Code :** 1000 1001 0111

8-3=5 **9-3=6** **7-3=4**

Decimal digits : 564

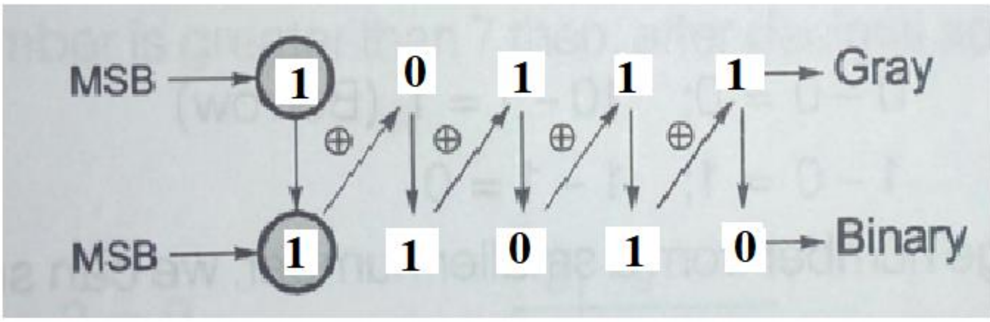
(ii)

(a) $(10010)_2 = (\ ?\ )_{Gray}$



Final result : $(10010)_2 = (11011)_{Gray}$

(b) $(10111)_{Gray} = (?)_2$



Final result : $(10111)_{Gray} = (11010)_2$

Q6 (i)

$$F = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}z + xy\bar{z} + xyz$$

$$= [\bar{x}\bar{y}z + x\bar{y}z] + \bar{x}y\bar{z} + xy[\bar{z} + z]$$

$$= [(\bar{x} + x)\bar{y}z] + \bar{x}y\bar{z} + xy\cdot 1$$

$$= 1\cdot\bar{y}z + [\bar{x}y\bar{z} + xy]$$

$$= \bar{y}z + [(\bar{x}\bar{z} + x)y]$$

$$= \bar{y}z + [(\bar{x} + x)(\bar{z} + x)y]$$

$$= \bar{y}z + [1\cdot(\bar{z}y + xy)]$$

$$= xy + y\bar{z} + \bar{y}z$$

(ii) $F(X,Y,Z) = \bar{X}Y\bar{Z} + X(\overline{YZ}) + XY\bar{Z} + XYZ$

$$= \bar{X}Y\bar{Z} + X(\bar{Y} + \bar{Z}) + XY\bar{Z} + XYZ$$

$$= \bar{X}Y\bar{Z} + X\bar{Y}\cdot 1 + X1\bar{Z} + XY\bar{Z} + XYZ$$

$$= \bar{X}Y\bar{Z} + X\bar{Y}(Z + \bar{Z}) + X(Y + \bar{Y})\bar{Z} + XY\bar{Z} + XYZ$$

$$F(X,Y,Z) = \Sigma(2,4,5,6,7)$$

$$= \Pi(0,1,3)$$

$$= (X + Y + Z)(X + Y + \bar{Z})(X + \bar{Y} + \bar{Z})$$